Supplementary Material for "Typed Decision Models as Passage Rerankers: Effectiveness, Calibration, and Context Selection"

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Additional Results

Table S1 reports effectiveness as a function of reranking depth; Table S2 reports both probability readouts for every collection with judged negatives; Table S3 reports pool coverage and the pool-bias check for calibration; Table S4 reports generator-free selection quality for RQ3; Table S5 reports exact match for the RAG policies of Table 6 of the article; Table S9 reports the family-clustered F1 and exact-match tests on the family-disjoint MuSiQue split; Table S10 reports the held-out calibration of the RQ3 probabilities. Code, prompts, and aggregate results are available in the reproduction repository (link withheld for anonymous review). All inputs are public and are downloaded by the code; the RAG splits are regenerated from a fixed seed, and the scripts reproduce every table and figure from the model outputs.

RAG prompt

System message: "Answer the question. Give only the short answer (a few words), with no explanation." User message: "Answer the question based on the given passages. Give only the short answer (a few words), with no explanation." followed by a blank line, the selected passages in rank order (each as "Passage i (Title: …): …", separated by blank lines), a blank line, and "Question: {question}\nAnswer:". When a policy selects no passage, the user message is only "Question: {question}\nAnswer:".

Yes/no prompt

System: "You are a search relevance assessor." User: "Query: {query}\nPassage: {passage}\n\nDoes the passage answer the query? Answer with Yes or No only."

RQ3 details

Generators were called through OpenRouter on 2026-10-04 by model name (`qwen/qwen-2.5-7b-instruct`, `meta-llama/llama-3.1-8b-instruct`, `qwen/qwen3-30b-a3b-instruct-2507`); OpenRouter does not expose weight snapshots. Qwen2.5-7B was served by one provider (Phala); Llama-3.1-8B by DeepInfra (61–64% of calls), Novita (25–28%), Groq (10%), and Cloudflare (1%); Qwen3-30B-A3B by six providers (StreamLake 31–32%, SiliconFlow 24%, Nebius 21–22%, Alibaba 11%, PrimeIntellect 11%, DekaLLM 1–2%). Jev's probability readout was chosen by Brier score on the calibration split (MuSiQue: Choice-3 strict 0.110, lenient 0.099, Noul 0.094 → Noul; NQ: 0.122, 0.286, 0.143 → Choice-3 strict; family-disjoint MuSiQue calibration split: 0.081, 0.080, 0.068 → Noul). Platt parameters (a, b) for $\sigma(a\,\mathrm{logit}(p) + b)$: Jev $(0.93, 0.60)$ and bge-reranker-v2-m3 $(0.42, 0.00)$ on MuSiQue; Jev $(1.01, 0.35)$ and bge-reranker-v2-m3 $(0.49, -0.01)$ on the family-disjoint MuSiQue split; Jev $(0.17, -0.77)$ and bge-reranker-v2-m3 $(0.34, -1.75)$ on NQ. Family-disjoint generations were run on 2026-10-17 with the same generator model names and prompt. Jev was called once per pair with default decoding. Its outputs are not fully deterministic: re-scoring DL19 and SciFact with the same snapshot in all three formats left 25–86% of pairs identical, changed probabilities by 0.0004–0.010 on average (at most 0.21), and gave a mean per-query Kendall τ of 0.92–0.96 between runs; nDCG@10 changed by at most 0.9 points on DL19 and 0.3 on SciFact. Complete SHA-256 hashes for the MuSiQue and DPR input files are recorded in the reproduction manifest.

Raw-probability thresholds (post-hoc)

These avoid Platt fitting but retain the readout chosen by Brier score on labelled calibration questions. At $t = 0.3/0.5/0.7$, Jev selects on average 1.4/0.8/0.4 passages on MuSiQue (empty for 25/46/70%) and 3.1/2.5/1.9 on NQ (19/25/33%). On MuSiQue, raw $P > 0.3$ is 1.6–2.4 F1 below the Platt threshold and 3.4–5.9 below top-3 (all generators, $p < 0.02$ except Llama versus Platt, $p = 0.07$); on NQ it is within 1.3 F1 of both (only Qwen2.5-7B versus Platt is significant, $+0.8$). These are not label-free policies.

Pooled-inference sensitivity

Table S7 contrasts normal and modified Knapp–Hartung intervals using the same DL heterogeneity estimates. The CSV additionally covers the DL and Clef–Jev format comparisons. Jev S4 versus bge on DL has mKH 90% CI $[-1.81, 3.16]$: non-inferiority at -2 survives, while equivalence does not. Clef–Jev equivalence survives for all three formats. Jev S4 versus Qwen8B and Clef S4 versus its base do not establish ± 2 equivalence under mKH. These analyses and margins are exploratory. Random-effects 95% prediction intervals for the difference on a new

Table S1. nDCG@10 when only the BM25 top-k is reranked (simulated exactly from the pointwise scores; passages below rank k keep their BM25 order). Averages over TREC DL19–23 and BEIR-8. The cost of a pointwise reranker grows linearly with k. BM25 alone: DL 39.3, BEIR 43.6.

	DL19–23					BEIR-8				
System	k=10	k=20	k=30	k=50	k=100	10	20	30	50	100
MiniLM	43.3	50.8	53.8	56.7	59.6	44.9	46.5	46.6	46.8	47.1
bge-v2-m3	43.7	51.6	54.6	57.8	60.6	45.9	48.4	48.7	49.3	49.3
Qwen3-Rer.-0.6B	42.9	49.5	52.0	54.1	56.7	45.7	48.4	49.2	50.0	50.5
Laya C3	41.8	46.2	46.9	48.0	47.3	43.7	43.1	41.5	39.6	37.4
Jeff Noul	41.9	46.1	46.3	46.7	45.1	43.1	42.3	40.5	38.8	35.9
Jev S4	44.0	51.9	55.1	58.0	61.1	46.8	50.4	51.4	52.5	53.0
BEIR-8: condensed nDCG′@10 (left), Judged@10 in % (right)										
MiniLM	60.8	62.3	62.9	63.6	64.2	57	58	57	56	56
bge-v2-m3	61.2	63.4	64.1	65.1	65.7	57	57	57	56	55
Qwen3-Rer.-0.6B	61.4	63.8	64.9	66.0	66.9	57	57	57	57	56
Laya C3	60.4	61.5	61.7	61.9	61.9	57	55	53	51	48
Jeff Noul	60.2	61.1	61.0	61.3	60.8	57	55	53	51	48
Jev S4	61.6	64.8	66.2	67.7	69.1	57	58	57	56	56

Table S2. ECE$\times$100 of Choice-3/Score-4 probabilities under the strict (s) and lenient (l) readouts. The readout used in the main text is underlined. Noul has a single readout.

System		DL19	DL20	DL21	DL22	DL23	Covid	DBP	News	Rob04
Laya C3	s	14.1	6.4	7.3	8.3	12.1	25.4	16.1	14.3	5.6
	l	24.5	34.0	36.6	42.9	41.9	6.9	16.7	22.0	30.4
Jeff C3	s	7.1	15.9	12.3	17.6	17.3	25.9	3.3	14.9	11.0
	l	36.3	45.3	42.8	48.6	47.7	11.6	21.0	23.9	48.2
Jev C3	s	12.9	9.3	13.1	8.2	14.3	43.9	26.7	30.6	14.2
	l	37.7	47.4	47.6	48.2	52.4	15.8	22.6	23.2	18.6
Jev S4	s	15.8	23.5	24.8	20.1	26.7	13.8	9.7	19.5	7.2
	l	52.8	68.0	59.8	70.3	69.3	27.4	42.6	40.6	32.3

Table S3. Pool coverage and pool-bias checks. Judged@10 (%) and condensed (judged-only) nDCG′@10, averaged over the collections each system was run on; ECE$\times$100 over the nine dense collections on judged pairs only vs. treating unjudged candidates as non-relevant; mean $P(\text{rel})$ on judged relevant / non-relevant pairs. *Run on a subset of collections (Clef: 7; Qwen3-235B: 8), so its averages are not directly comparable.

	Judged@10		nDCG′	ECE		mean P	
System	DL	BEIR	All	judged	unj.=0	rel.	non-rel.
BM25	99.9	56.7	51.8	–	–	–	–
MiniLM	96.9	55.7	62.6	35.4	30.6	0.77	0.49
bge-v2-m3	95.9	55.5	64.0	20.0	15.2	0.64	0.28
Qwen3-Rer.-0.6B	93.4	56.4	63.5	36.1	31.5	0.83	0.51
Qwen3-235B P(yes)*	87.1	74.9	62.9	20.5	15.8	0.74	0.24
Laya C3	81.1	48.1	58.6	11.0	14.2	0.50	0.30
Jeff C3	76.4	48.1	57.3	15.3	23.5	0.57	0.41
Clef C3*	90.5	97.1	66.9	7.8	6.8	0.50	0.18
Clef S4*	91.5	97.4	67.5	11.6	14.0	0.66	0.30
Jev C3	89.7	55.4	66.3	14.8	13.3	0.62	0.23
Jev Noul	89.8	55.3	66.7	15.0	12.7	0.67	0.26
Jev S4	90.0	55.7	66.9	25.4	25.1	0.86	0.43

Table S4. Generator-free selection quality on the RQ3 final sets: recall of gold passages within the top-k (all-gold: every gold passage selected), and, on MuSiQue, whether the top-1 passage is the first-hop or the last (answer-bearing) supporting paragraph and whether it contains the answer string.

Data	Ranker	R@1	R@3	R@5	all-gold@5
MuSiQue	Jev	33.3	68.7	83.2	61.4
MuSiQue	bge-v2-m3	35.4	66.0	76.9	49.5
MuSiQue	Qwen3-Reranker-8B	34.9	68.6	78.7	51.5
MuSiQue	Qwen3-Reranker-0.6B	33.7	62.1	71.5	38.8
NQ	Jev	27.3	58.4	72.2	47.4
NQ	bge-v2-m3	26.1	52.5	65.8	39.1
NQ	Qwen3-Reranker-8B	25.6	55.1	69.0	43.0
NQ	Qwen3-Reranker-0.6B	21.4	47.1	61.6	34.8
MuSiQue top-1:	first hop / last hop / contains answer (%)				
	Jev	22.6 / 51.2 / 56.5			
	bge-v2-m3	61.9 / 15.2 / 22.3			
	Qwen3-Reranker-8B	44.5 / 30.2 / 36.6			
	Qwen3-Reranker-0.6B	56.0 / 16.5 / 23.0			

collection ($\hat{\mu} \pm t_{k-2}\sqrt{\hat{\tau}^2 + \mathrm{SE}^2}$) are wide: $[-6.9, 7.5]$ for Jev S4 minus Qwen8B, $[-7.5, 12.6]$ for Jev S4 minus bge ($[-4.0, 7.4]$ without Robust04), and $[-3.7, 5.7]$ for Clef S4 minus its base.

Table S5. RQ3: exact match (EM) on the same questions, policies, and generations as Table 6 of the article. Avg. n: mean number of selected passages. †: Jev significantly better than bge-reranker-v2-m3 under the same policy on EM (question-level paired t-test, Holm-corrected over the nine policies, $p < 0.05$); these tests do not account for MuSiQue family dependence. EM is a secondary metric.

	MuSiQue			NQ			Avg. n	
Policy	Qwen2.5-7B	Llama-3.1	Qwen3-30B	Qwen2.5-7B	Llama-3.1	Qwen3-30B	MuSiQue	NQ
No context	2.9	3.5	4.3	16.1	19.9	23.0	0.0	0.0
All 20 passages	27.8	27.0	34.9	40.6	38.4	37.4	20.0	20.0
Gold passages (oracle)	47.7	41.6	51.6	47.7	47.5	45.0	2.6	3.4
Jev 1.13								
top-1	34.4†	26.1†	33.1†	38.5	38.5	38.8	1.0	1.0
top-3	32.6†	29.0†	36.7†	39.7	38.2	39.0	3.0	3.0
top-5	31.2	30.8†	36.5†	39.0	39.2	39.2	5.0	5.0
adaptive-k	34.8†	29.3†	36.5†	39.8	38.6	38.1	2.9	6.2
$P > 0.3$	32.6†	26.1†	32.5†	38.8	39.9	38.3	2.4	2.9
top-2 + $P > 0.3$	32.7†	28.6†	34.7†	39.6	39.5	39.4	3.0	3.5
bge-reranker-v2-m3								
top-1	12.7	10.2	13.7	36.7	36.8	37.1	1.0	1.0
top-3	27.2	22.3	29.1	38.5	39.0	38.1	3.0	3.0
adaptive-k	20.2	16.4	22.4	38.5	39.3	38.5	3.1	7.1
top-2 + $P > 0.3$	24.9	21.1	27.6	38.3	39.9	38.5	2.9	3.9
Qwen3-Reranker-8B (4-bit)								
top-1	20.3	15.9	21.5	35.0	33.9	35.2	1.0	1.0
top-3	29.9	24.4	33.5	37.3	39.3	37.8	3.0	3.0
adaptive-k	26.6	23.0	27.9	39.0	38.5	37.2	3.7	10.3
top-2 + $P > 0.3$	27.9	25.3	31.0	37.7	39.0	36.9	3.1	4.4
Qwen3-Reranker-0.6B								
top-1	13.4	10.0	13.3	30.6	31.6	32.2	1.0	1.0
top-3	24.6	20.4	26.0	36.5	36.2	35.2	3.0	3.0
adaptive-k	20.5	17.9	22.6	39.1	38.4	36.5	3.2	10.1
top-2 + $P > 0.3$	22.2	19.9	24.8	37.7	36.0	36.4	2.8	3.8

Table S6. Retrospective MuSiQue sensitivity on 329 questions with no sub-question ID in calibration or pilot. Answer F1 under three ranker-effect policies; Δ is Jev minus bge. CIs resample 212 source-graph families (20,000 paired replicates, seed 13), retaining question weights; intervals are unadjusted. †: passes the original nine-policy question-level Holm correction (top-1 and Adaptive-k for every generator; no top-3 comparison).

Generator	Policy	Jev	bge	Δ	Family 95% CI
Qwen2.5-7B	top-1	41.8	23.1	+18.8†	[14.0, 23.8]
	Adaptive-k	42.1	31.3	+10.7†	[5.9, 15.7]
	top-3	43.6	41.3	+2.4	[−2.6, 7.3]
Llama-3.1-8B	top-1	35.0	20.6	+14.4†	[9.6, 19.2]
	Adaptive-k	38.4	27.1	+11.3†	[7.1, 15.7]
	top-3	41.1	35.7	+5.5	[1.0, 10.0]
Qwen3-30B-A3B	top-1	43.9	26.0	+17.9†	[12.5, 23.4]
	Adaptive-k	47.9	34.6	+13.3†	[9.0, 17.8]
	top-3	48.1	42.4	+5.7	[0.8, 10.5]

Prompt sensitivity

We fixed three paraphrases each of the Noul question, the Qwen3-235B yes/no prompt, and the Qwen3-Reranker instruction before running any of them (code: `PARAPHRASES`, `USER_PARAPHRASES`, `INSTRUCTIONS`). For Qwen3-Reranker-0.6B on DL19, DL20, and SciFact, the paraphrases change nDCG@10 by at most 2.0 points (DL19 69.4 with the default instruction versus 71.2–71.4), but they lower raw ECE on DL from 39.0 and 46.2 to 23.5–26.0 and 28.8–32.5; on SciFact ECE stays within 3.0–4.4. The default instruction is therefore the least calibrated of the four wordings, so raw-ECE comparisons with the Qwen rerankers depend on the prompt. Jev's Noul question is equally robust in ranking (nDCG@10 within 2.1 points of the original on all three collections) but not in calibration: raw ECE ranges from 7.5 to 25.9 on DL19 (original 12.4), 14.3 to 34.5 on DL20 (20.7), and 2.0 to 6.5 on SciFact (3.1). The original wording sits in the middle of this range, and the most direct wording ("Is the passage relevant to the search query?") is the least calibrated on every collection. Raw-ECE differences of a few points between Jev and the baselines (Table 4 of the article) are therefore within the spread that rewording alone produces. For Qwen3-235B, the paraphrases also keep nDCG@10 within 2.3 points (DL19 68.6 versus 69.8–70.9; DL20 66.5 versus 65.5–66.6), but the original wording is the best calibrated: raw ECE rises from 18.6 to 20.6–30.4 on DL19 and from 22.0 to 25.3–34.9 on DL20. On SciFact (no original run) the paraphrases agree within 0.3 nDCG@10 and 1.6 ECE points. Paraphrase 3 and the last 6% of paraphrase 2 were served by one deployment, after the other retired the model.

Neural first stage

To check whether the system ordering depends on BM25, we reranked RankLLM's SPLADE++ EnsembleDistil[1] top-100 candidates for DL19 and DL20 (same qrels) with bge-reranker-v2-m3, MiniLM, Qwen3-Reranker-8B, and Jev's three formats. SPLADE++ alone reaches 73.1/72.0 nDCG@10 on DL19/DL20 (BM25: 50.6/48.0), so every reranker gains much less. bge-reranker-v2-m3 scores 76.0/75.6, Qwen8B 75.5/73.1, and MiniLM 74.9/74.4. Jev scores 70.5–72.9 on DL19, 3.1–5.5 points below bge (Score-4: $p = 0.03$, unadjusted) and below the SPLADE++ ranking itself, whereas on BM25 candidates it was within 1.1 points of bge. On DL20 it scores 74.6–77.1, and Noul is the best system (77.1, +1.5 over bge, $p = 0.50$). Jev, however,

Table S7. Post-hoc sensitivity of pooled ranking inference to small-sample uncertainty. Both methods use the same DerSimonian–Laird heterogeneity estimate; mKH uses a Student-t reference with $k-1$ degrees of freedom and a residual variance multiplier floored at one. Units: nDCG@10 points. A 90% interval inside ± 2 corresponds to the exploratory equivalence criterion; the margin was chosen after observing results.

Comparison	k	Pooled difference	Normal CI	mKH CI
Jev S4 – bge, all (95%)	13	+2.60	[0.07, 5.13]	[-0.37, 5.57]
Jev S4 – bge, no Robust04 (95%)	12	+1.73	[0.09, 3.36]	[-0.42, 3.87]
Jev S4 – Qwen8B, all (90%)	13	+0.31	[-1.26, 1.89]	[-1.39, 2.02]
Clef S4 – base, seven collections (90%)	7	+0.98	[-0.27, 2.23]	[-0.50, 2.46]
Jev S4 – Qwen8B, no Robust04 (90%)	12	-0.29	[-1.36, 0.77]	[-1.59, 1.00]

Table S8. nDCG@10 for reranking the BM25 top-100 (Pyserini runs and qrels, as in RankGPT/RankLLM). $^{\uparrow}/^{\downarrow}$: significantly better/worse than bge-reranker-v2-m3 (paired t-test over queries, Holm-corrected over the collections each pair was evaluated on, $p < 0.05$). Best per column in bold. Averages are reported only for systems run on every collection of the group (dash otherwise); matched comparisons for Clef and the P(yes) LLMs are given in the text. Touche, News and Robust04 contain long documents that every model truncates (Table 3 of the article).

	TREC Deep Learning						BEIR									
System	DL19	DL20	DL21	DL22	DL23	*Avg*	Covid	NFC	Touche	DBP	SciF	Signal	News	Rob04	*Avg*	**All**
BM25 (first stage)	50.6	48.0	44.6	26.9	26.2	39.3	59.5	32.2	**44.2**	31.8	67.9	33.1	39.5	40.7	43.6	41.9
Cross-encoders																
MiniLM-L6 (22M)	72.7	67.5	65.6	49.9	42.2	59.6	74.5	34.9	28.4$^{\downarrow}$	43.9	69.5$^{\downarrow}$	**33.9**$^{\uparrow}$	44.3	47.3$^{\downarrow}$	47.1	51.9
bge-reranker-v2-m3 (568M)	72.4	68.1	66.2	**51.4**	45.1	60.6	78.3	34.3	38.3	44.0	75.0	29.0	43.6	51.9	49.3	53.7
Label-probability LLM rerankers																
Qwen3-Reranker-0.6B	69.4	64.0	65.1	46.3$^{\downarrow}$	38.8$^{\downarrow}$	56.7	**86.2**$^{\uparrow}$	36.8$^{\uparrow}$	29.2$^{\downarrow}$	43.3	76.8	30.6	45.5	55.8$^{\uparrow}$	50.5	52.9
Qwen3-Reranker-4B (4-bit)	**74.0**	68.2	66.2	50.4	45.0	60.7	84.1$^{\uparrow}$	37.1$^{\uparrow}$	34.9	**46.4**$^{\uparrow}$	77.8	33.8$^{\uparrow}$	48.6$^{\uparrow}$	57.4$^{\uparrow}$	52.5	55.7
Qwen3-Reranker-8B (4-bit)	73.6	66.5	66.2	51.1	44.9	60.5	83.8$^{\uparrow}$	37.5$^{\uparrow}$	36.2	45.5	79.2$^{\uparrow}$	33.7$^{\uparrow}$	**51.6**$^{\uparrow}$	58.0$^{\uparrow}$	**53.2**	56.0
Qwen3-235B, P(yes)	68.6	66.5	65.6	46.2	42.8	57.9	80.0	34.6	–	–	–	–	–	60.2$^{\uparrow}$	–	–
Qwen3-235B, graded 0–4	71.6	66.3	68.5	45.5$^{\downarrow}$	44.1	59.2	83.0	37.5$^{\uparrow}$	–	–	–	–	–	61.9$^{\uparrow}$	–	–
Qwen3.8-27B (Clef's base), P(yes)	73.0	69.8	69.2	48.8	45.5	61.2	83.7	–	–	–	–	–	–	62.5$^{\uparrow}$	–	–
Qwen3.5-0.8B (Jeff's base), P(yes)	42.9$^{\downarrow}$	38.9$^{\downarrow}$	50.6$^{\downarrow}$	26.3$^{\downarrow}$	28.6$^{\downarrow}$	37.4	60.2$^{\downarrow}$	17.3$^{\downarrow}$	19.1$^{\downarrow}$	21.4$^{\downarrow}$	28.0$^{\downarrow}$	24.4$^{\downarrow}$	22.7$^{\downarrow}$	33.5$^{\downarrow}$	28.3	31.8
Qwen3.5-0.8B (Jeff's base), graded 0–4	34.8$^{\downarrow}$	31.7$^{\downarrow}$	33.5$^{\downarrow}$	18.7$^{\downarrow}$	20.6$^{\downarrow}$	27.9	60.0$^{\downarrow}$	17.1$^{\downarrow}$	16.7$^{\downarrow}$	20.1$^{\downarrow}$	28.0$^{\downarrow}$	14.2$^{\downarrow}$	22.1$^{\downarrow}$	25.1$^{\downarrow}$	25.4	26.4
Laya (421M)																
Choice-3	60.0$^{\downarrow}$	52.1$^{\downarrow}$	54.7$^{\downarrow}$	36.0$^{\downarrow}$	33.8$^{\downarrow}$	47.3	74.1	29.3$^{\downarrow}$	23.3$^{\downarrow}$	29.6$^{\downarrow}$	48.6$^{\downarrow}$	22.6$^{\downarrow}$	37.2$^{\downarrow}$	34.6$^{\downarrow}$	37.4	41.2
Noul	57.1$^{\downarrow}$	52.1$^{\downarrow}$	55.0$^{\downarrow}$	36.2$^{\downarrow}$	32.0$^{\downarrow}$	46.5	74.2	28.9$^{\downarrow}$	22.5$^{\downarrow}$	28.7$^{\downarrow}$	44.5$^{\downarrow}$	21.8$^{\downarrow}$	35.7$^{\downarrow}$	34.5$^{\downarrow}$	36.4	40.3
Score-4	62.2$^{\downarrow}$	53.0$^{\downarrow}$	53.0$^{\downarrow}$	37.5$^{\downarrow}$	33.4$^{\downarrow}$	47.8	74.1	27.2$^{\downarrow}$	23.5$^{\downarrow}$	25.9$^{\downarrow}$	39.4$^{\downarrow}$	21.3$^{\downarrow}$	36.0$^{\downarrow}$	32.8$^{\downarrow}$	35.0	40.0
Jeff (0.8B)																
Choice-3	55.4$^{\downarrow}$	46.1$^{\downarrow}$	49.0$^{\downarrow}$	32.1$^{\downarrow}$	28.9$^{\downarrow}$	42.3	68.7$^{\downarrow}$	24.6$^{\downarrow}$	22.9$^{\downarrow}$	29.3$^{\downarrow}$	54.4$^{\downarrow}$	27.0	34.2$^{\downarrow}$	35.8$^{\downarrow}$	37.1	39.1
Noul	59.5$^{\downarrow}$	50.1$^{\downarrow}$	52.5$^{\downarrow}$	33.0$^{\downarrow}$	30.4$^{\downarrow}$	45.1	67.3$^{\downarrow}$	24.5$^{\downarrow}$	23.5$^{\downarrow}$	30.0$^{\downarrow}$	46.7$^{\downarrow}$	27.4	33.6$^{\downarrow}$	34.1$^{\downarrow}$	35.9	39.4
Score-4	55.1$^{\downarrow}$	47.1$^{\downarrow}$	50.0$^{\downarrow}$	32.4$^{\downarrow}$	30.1$^{\downarrow}$	42.9	67.9$^{\downarrow}$	24.6$^{\downarrow}$	23.4$^{\downarrow}$	27.4$^{\downarrow}$	55.0$^{\downarrow}$	27.0	34.5$^{\downarrow}$	35.6$^{\downarrow}$	36.9	39.2
Clef 27B (open weights, API; 7 collections)																
Choice-3	71.5	70.0	66.8	48.5	44.7	60.3	82.8	–	–	–	–	–	–	66.5$^{\uparrow}$	–	–
Noul	71.3	69.9	67.3	47.7	46.4	60.5	83.1	–	–	–	–	–	–	66.2$^{\uparrow}$	–	–
Score-4	72.2	**70.2**	69.3	49.1	**46.8**	**61.5**	84.0	–	–	–	–	–	–	**66.5**$^{\uparrow}$	–	–
Jev 1.13 (API)																
Choice-3	71.7	70.0	68.5	46.8	43.1	60.0	82.8	37.9$^{\uparrow}$	32.1	43.2	79.6$^{\uparrow}$	30.3	46.6	66.0$^{\uparrow}$	52.3	55.3
Noul	71.3	70.0	69.5	50.1	45.0	61.2	84.1	37.7$^{\uparrow}$	35.5	42.7	79.8$^{\uparrow}$	28.3	47.6	66.0$^{\uparrow}$	52.7	56.0
Score-4	72.8	69.9	**70.0**$^{\uparrow}$	48.5	44.3	61.1	85.0$^{\uparrow}$	**38.4**$^{\uparrow}$	31.6	44.5	**80.0**$^{\uparrow}$	29.8	48.6	65.9$^{\uparrow}$	53.0	**56.1**

Table S9. Family-disjoint MuSiQue (597 questions in 270 families): Jev minus bge-reranker-v2-m3 and minus Qwen3-Reranker-8B under each policy, in answer F1 and exact match (EM). *: significant in the family cluster-robust paired t-test (CR1, 269 degrees of freedom), Holm-corrected over the nine policies within each generator, comparison, and metric; this is the primary MuSiQue test. Question-level tests, family sign-flip tests, and family-bootstrap intervals for every cell, and the same tests on the original split, are in the reproduction repository.

	Jev − bge-reranker-v2-m3						Jev − Qwen3-Reranker-8B					
	F1			EM			F1			EM		
Policy	Qw2.5	Llama	Qw3	Qw2.5	Llama	Qw3	Qw2.5	Llama	Qw3	Qw2.5	Llama	Qw3
top-1	+16.9*	+12.1*	+13.3*	+15.1*	+9.4*	+12.2*	+9.7*	+6.4*	+6.4*	+8.7*	+4.9*	+5.7*
top-2	+7.2*	+10.2*	+8.1*	+5.9	+7.9*	+6.9*	+4.5*	+4.7	+5.1*	+3.5	+3.5	+3.2
top-3	+5.2	+7.8*	+5.8*	+3.9	+6.9*	+6.5*	+5.7*	+4.3	+4.9	+3.4	+3.0	+4.9
top-5	+3.6	+3.5*	+2.8	+4.4	+4.4*	+2.7	+2.8	+2.0	+2.1	+3.4	+2.0	+1.8
adaptive-k	+12.6*	+11.8*	+12.7*	+11.6*	+9.5*	+12.7*	+7.4*	+5.1*	+7.0*	+6.5*	+4.2	+6.7*
top-2 + $P > 0.3$	+7.7*	+8.7*	+6.7*	+5.9	+6.9*	+6.0*	+4.6*	+3.4	+2.6	+2.7	+2.7	+1.7
top-2 + $P > 0.5$	+7.3*	+9.7*	+7.7*	+6.0	+7.5*	+6.5*	+5.3*	+4.1	+5.1*	+4.0	+2.7	+2.8
$P > 0.2$	+6.8*	+4.9*	+4.3	+5.5	+4.2*	+2.8	+3.2	+0.6	−0.2	+2.5	+0.8	−1.0
$P > 0.3$	+7.6*	+7.1*	+6.4*	+6.9*	+6.2*	+5.2	+3.6	+3.0	+0.1	+2.8	+3.0	−0.7
Significant	23/27			18/27			12/27			5/27		

Table S10. Held-out calibration of the RQ3 probabilities on the final candidate pairs (ECE×100, 15 bins; gold supporting or answer-bearing passage as the label). Raw: as returned (cross-encoder logits through a sigmoid). Platt: fit on the calibration split of the same dataset and applied to the final split, as for the threshold policies. Underlined: the Jev readout used for thresholds (chosen by Brier score on the calibration split). FD: family-disjoint MuSiQue (11,939 pairs); Orig.: original MuSiQue final split (19,988); NQ: 20,000. Base rate of relevant pairs: 0.107, 0.133, 0.171.

	MuSiQue FD		MuSiQue Orig.		NQ	
Ranker	Raw	Platt	Raw	Platt	Raw	Platt
Jev Noul	<u>3.1</u>	<u>3.1</u>	<u>5.1</u>	<u>3.4</u>	10.2	1.2
Jev C3 (strict)	7.4	2.1	10.2	2.3	<u>9.5</u>	<u>3.8</u>
Jev S4 ($\ell \geq 2$)	3.4	1.5	4.8	2.8	15.0	3.7
bge-reranker-v2-m3	4.7	1.3	6.7	1.4	25.3	1.4
Qwen3-Rer.-8B (4-bit)	5.0	1.4	6.6	1.2	17.4	1.8
Qwen3-Rer.-0.6B	8.1	1.8	9.3	1.6	33.9	1.5

places more unjudged candidates in its top 10 than the cross-encoders do (Judged@10 86.7–87.9% on DL19 versus 94.9% for bge and 94.2–94.7% for Qwen8B and MiniLM; 90.9–91.9% versus 97.6% for bge on DL20), and standard nDCG@10 counts them as non-relevant. On condensed lists, which drop unjudged candidates before scoring[2], Jev's DL19 gap to bge shrinks to -0.1 to -1.7 points (none significant), Jev is no longer below the SPLADE++ ranking (76.0–77.5 versus 75.9), and on DL20 it leads bge by $+3.0$ to $+4.0$ (Noul $p = 0.04$, unadjusted). Neither scoring bounds the difference under complete judgments: standard scoring counts unjudged candidates as non-relevant and condensed scoring removes them, but an unjudged candidate could carry any grade. Setting every unjudged candidate in the two top-10 lists (1.8 per query on average) to grade 1, 2, or 3 gives a Jev S4 minus bge difference of -1.8, $+1.5$, or $+4.0$ on DL19, and the exact range over all completions is $[-8.7, +7.8]$. The DL19 deficit therefore cannot be separated from incomplete judgments, and even its sign is not established. The BM25 pools are also less complete for Jev (Judged@10 90.5–93.5% versus 96.0–97.2% for bge); there, condensed scoring moves Jev's DL19 difference to bge from -1.0 to $+0.5$ to between $+0.9$ and $+1.8$ points (not significant).

Split and provenance audit

Calibration, pilot, and final question IDs are disjoint, but MuSiQue families overlap substantially. Comparing `question_decomposition[-1].id`, 494 final questions share the terminal sub-question with pilot, 349 with calibration, 265 with both, and 578 with either. Comparing any decomposition ID gives 593 pilot overlaps, 430 calibration overlaps, and 671 with either (352 with both). Counts are affected final questions, not unique IDs or matching pairs. The earlier whole-question normalization check found one pilot/final duplicate but missed component reuse. The local component audit checks the source hash and all six cached ranker split-ID sets; its CSV records question-level memberships. The 329-question sensitivity excludes any directly shared component, yet 23 questions in 18 source-graph families remain indirectly connected to design questions. It therefore cannot establish fully family-disjoint generalization. The audit and sensitivity scripts preserve the original splits and model outputs. The reproduction manifest records complete input SHA-256 hashes, all split IDs, installed package versions, analysis-source hashes, and the run-manifest hash, and links the component audit. Historical Hugging Face weight commits were not recorded; architecture strings cannot recover exact weight revisions. Original cached outputs should be preserved for historical numerical verification.

References

1. Formal T, Lassance C, Piwowarski B et al. From distillation to hard negative sampling: Making sparse neural IR models more effective. In *Proceedings of SIGIR 2022*. pp. 2353–2359. DOI: 10.1145/3477495.3531857.
2. Sakai T. Alternatives to Bpref. In *Proceedings of the 30th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR 2007)*. pp. 71–78. DOI:10.1145/1277741.1277756.